

1. AI Tools & Models Used Primary LLM: Gemini (via Google) for video information retrieval, transcript analysis, and slide layout structuring.

Secondary Generation Tool: [Insert Gamma.app, ChatGPT, or PowerPoint here, depending on what you use to render the slides, for converting text markdown into a visual presentation deck.

2. The Complete Process. The objective of this assignment was to leverage Large Language Models to distill a 49-minute biographical lecture video into a concise, professional 10-slide presentation deck while documenting the iterative prompting process. The workflow was executed in four distinct phases:

Phase 1: Content Ingestion & Token Management: Rather than watching the video linearly, the YouTube URL (https://www.youtube.com/watch?v=fNh_GyRbJDA) was fed into the LLM via specialized tools to extract the complete, timestamped transcript. This bypassed manual note-taking and prevented information gaps.

Phase 2: Chronological & Thematic Chunking: The massive raw transcript text was analyzed by the LLM to identify structural pivots in Rockefeller's life. The data was organized into 5 core historical phases: Early Life/Poverty, Rise of Standard Oil, Monopolistic Dominance, Federal Antitrust Warfare, and Systematic Philanthropy.

Phase 3: Slide Deck Outlining: A systematic prompting strategy was used to compress paragraphs of historical data into slide-ready bullet points, ensuring key terminology (e.g., horizontal integration, railroad rebates, drawbacks) was properly defined and placed.

Phase 4: Presentation Rendering & Report Compilation: The optimized text was exported to a presentation generator to finalize visual assets, layouts, and speaker notes, while the prompt history was compiled into this analysis report.

3. Prompt Log & Analysis Original / Initial Prompts Used Prompt 1 (Exploratory): "Read this video transcript about John D. Rockefeller and write a 10-slide presentation deck summarizing his entire life."

What Worked Effectively. The initial prompt successfully extracted the primary timeline, but the slides were incredibly text-heavy. To fix this, strict constraint-based prompting was used.

The Winning Prompt Structure: "Act as an expert corporate presentation designer. Create a slide layout for the 'Cleveland Massacre'. Limit the slide to exactly 3 bullet points. Each bullet point must be under 15 words. Include one historical quote and a suggestion for a visual asset."

Why it worked: Forcing the LLM into a specific persona (Presentation Designer) and setting hard numerical constraints (3 bullets, under 15 words) prevented the model from dumping paragraphs of text onto a single slide.

What Did Not Work Well: The Failure Scenario: When asking the model to handle complex financial concepts broadly, it failed to capture the nuances of his business tactics. For instance, asking it to "Explain how Rockefeller beat the railroads" resulted in a generic essay that confused his horizontal integration with his vertical integration.

The Timeline Mashup: The model initially placed his rivalry with Andrew Carnegie after the dissolution of Standard Oil, which is historically inaccurate. The LLM also kept defaulting to cliché corporate filler phrasing (e.g., "Rockefeller was a man of vision who changed the world") instead of sticking to hard historical data and specific events.

Iterative Refinement Process. To correct the inaccuracies and poor slide formatting, the prompting method was shifted from a "single-shot request" to an Iterative Chunking Strategy.

Instead of asking for all 10 slides at once, the model was prompted slide-by-slide. For Slide 4 (Predatory Practices), the prompt was refined to: "Focus exclusively on the mechanics of railroad rebates. Do not talk about his childhood or his death. Explain how lower transportation costs directly funded his price wars." This exact guidance produced clean, accurate, and scannable content.